Pandrosion Geometry in Several Complex Variables

Integral slopes, homothetic transport, guarded continuation,
and the resultant–projective completion architecture of Flow 092

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Abstract

We formulate the multidimensional Pandrosion construction used in the implementation `flow/092_pandrosion_resultant_pairing.py`. The setting is a square polynomial system

$$F = (F_1, \ldots, F_n) : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

with isolated regular zeros. The basic object is not a derivative at a point, but a two-point slope matrix $Q(a, z)$ satisfying the exact telescoping identity

$$F(z) - F(a) = Q(a, z)(z - a).$$

The canonical geometric representative is the segment-integral matrix

$$Q^{\mathrm{int}}_{ij}(a, z) = \int_0^1 \partial_j F_i\big(a + t(z - a)\big)\, \mathrm{d}t,$$

which extends Pandrosion's one-dimensional divided quotient to arbitrary dimension. We then describe the system-adaptive exactified slope, convex blends of exact geometries, diagonal homothetic transport, Aitken–Pandrosion T_2 acceleration, and the branch-safe continuation guard.

The resulting algorithm remains a homotopy method on $\mathbb{C}^n$, but the predictor–corrector step is derivative-free in the Pandrosion sense: every accepted candidate is corrected by a non-local slope matrix rather than by a Newton–Lairez corrector. Flow 092 adds three completion layers: coordinate resultants for bivariate systems, deterministic multivariate gamma sweeps, and projective/Riemann-sphere charts. Resultants and projective charts are used only as candidate generators; acceptance is always by Pandrosion polish on the original system. We record the mathematical identities behind the construction, state local fixed-point and transport propositions, and give an empirical summary of the current default configuration. No complexity-theoretic claim against the Beltran–Pardo or Lairez algorithms is made.

Implementation note. The paper describes the construction implemented by the chain

$$092 \to 091 \to 090 \to 089 \to 087 \to 081 \to 076 \to 070 \to 069 \to 068.$$

The numbering is repository-local. The mathematical object is the multidimensional Pandrosion slope geometry; the code numbers are included only to make the implementation auditable.

Contents

1 Introduction

Pandrosion's original construction for p-th roots can be written as a fixed-point iteration generated by a geometric sum. In one complex variable, the essential identity is the divided quotient

$$P(z) - P(a) = Q_P(a, z)(z - a), \qquad Q_P(a, z) = \frac{P(z) - P(a)}{z - a}.$$

If ζ is a zero of P, then

$$\zeta = a - Q_P(a, \zeta)^{-1} P(a),$$

which suggests a root-refinement map

$$z \longmapsto a - Q_P(a, z)^{-1} P(a).$$

This map is not Newton's method unless the anchor a is collapsed onto the current iterate. The non-locality of $Q_P(a, z)$ is the geometric content of Pandrosion: the step is built from a chord, not a tangent.

For a system $F : \mathbb{C}^n \to \mathbb{C}^n$, the scalar quotient is replaced by a slope matrix $Q_F(a, z)$. The present paper develops the form used in the 092 solver: a portfolio of exact two-point geometries, transported by a diagonal homothety and embedded in a total-degree homotopy. The root-finding problem is the standard one:

$$F(z) = 0, \qquad z \in \mathbb{C}^n.$$

All variables, coefficients, homotopy parameters, and candidate roots live over $\mathbb{C}$.

What is new in 092 relative to the earlier flows. The earlier flows establish the Pandrosion corrector and the homothetic geometry. Flow 092 adds a practical completion architecture:

- branch-safe path tracking with residual and predictor-compatibility guards;
- bivariate coordinate-resultant recovery when $n = 2$;
- deterministic multivariate completion sweeps when $n \geq 3$;
- projective/Riemann-sphere recovery charts as a last-resort tool;
- tight bivariate clustering, because valid roots can be closer than the generic 10^{-6} clustering radius.

The empirical lesson of the current implementation is that these layers are most effective in this order: homothetic Pandrosion tracking first, then resultant/projective recovery only for missing sheets.

2 Polynomial systems over $\mathbb{C}$

Let

$$F = (F_1, \ldots, F_n), \qquad F_i \in \mathbb{C}[z_1, \ldots, z_n],$$

be a square system. We write $a, z \in \mathbb{C}^n$, $\Delta = z - a$, and

$$J_F(x) = (\partial_j F_i(x))_{1 \leq i,j \leq n} .$$

The Bézout number of the system is

$$D = \prod_{i=1}^{n} d_i, \qquad d_i = \deg F_i.$$

For the families used in the experiments, D is the target number of isolated solutions counted with multiplicity.

Definition 2.1 (Pandrosion slope matrix)**.** A matrix-valued map $Q(a, z) \in \mathbb{C}^{n \times n}$ is a Pandrosion slope for F between a and z if

$$F(z) - F(a) = Q(a, z)(z - a). \tag{1}$$

We call (1) the telescoping identity.

The identity is the load-bearing invariant of the construction. Different geometries correspond to different choices of Q, but all admissible choices must satisfy (1).

Definition 2.2 (Pandrosion correction map)**.** Let $Q(a, z)$ be an invertible slope matrix. The associated anchored Pandrosion map is

$$\mathcal{P}_Q(a, z) = a - Q(a, z)^{-1} F(a). \tag{2}$$

Proposition 2.3 (Fixed-zero property)**.** *If ζ is a zero of F and $Q(a, \zeta)$ is invertible, then*

$$\mathcal{P}_Q(a, \zeta) = \zeta.$$

Proof. Since $F(\zeta) = 0$, the telescoping identity gives

$$-F(a) = Q(a, \zeta)(\zeta - a).$$

Multiplying by $Q(a, \zeta)^{-1}$ yields

$$-Q(a, \zeta)^{-1} F(a) = \zeta - a,$$

which is equivalent to $\mathcal{P}_Q(a, \zeta) = \zeta$. □

Remark 2.4. Proposition 2.3 is the multidimensional analogue of the classical one-dimensional Pandrosion identity. It explains why a two-point slope can act as a corrector: at an actual zero, the corrector fixes the zero exactly.

3 The integral geometry

The cleanest slope matrix is obtained by integrating the Jacobian along the straight segment from a to z.

Definition 3.1 (Segment-integral Pandrosion slope)**.** For $a, z \in \mathbb{C}^n$, define

$$Q_{ij}^{\text{int}}(a,z) = \int_0^1 \partial_j F_i\big(a + t(z-a)\big)\,\mathrm{d}t. \tag{3}$$

Equivalently,

$$Q^{\text{int}}(a,z) = \int_0^1 J_F(a + t(z-a))\,\mathrm{d}t.$$

Theorem 3.2 (Exact telescoping for Q^{int})**.** *For every polynomial map $F : \mathbb{C}^n \to \mathbb{C}^n$,*

$$F(z) - F(a) = Q^{\text{int}}(a,z)(z-a).$$

Proof. Let $\gamma(t) = a + t(z-a)$. For each component F_i,

$$\frac{\mathrm{d}}{\mathrm{d}t} F_i(\gamma(t)) = \sum_{j=1}^{n} \partial_j F_i(\gamma(t))(z_j - a_j).$$

Integrating from 0 to 1 gives

$$F_i(z) - F_i(a) = \sum_{j=1}^{n} \left(\int_0^1 \partial_j F_i(a + t(z-a))\,\mathrm{d}t \right)(z_j - a_j),$$

which is the i-th row of the matrix identity. □

Numerical realization. For polynomial systems, the integrand in (3) is polynomial in t. A Gauss–Legendre rule of sufficient order is exact. The implementation uses a capped exact/approximate Gauss–Legendre rule:

$$Q^{\text{GL}}(a,z) = \sum_{\ell=1}^{m} w_\ell J_F(a + t_\ell(z-a)).$$

When the cap is high enough relative to $\max_i d_i$, this equals Q^{int} up to roundoff.

Remark 3.3 (Classical interpretation). The matrix Q^{int} is a Schmidt-type divided-difference matrix for systems. The Pandrosion contribution here is not the existence of divided differences; it is the use of these matrices as geometric two-point correctors, combined with homothetic scaling and guarded continuation.

4 System-adaptive exactification

The integral geometry is robust but can be expensive. Flow 070 introduced a cheaper system-adaptive geometry. It begins from an edge slope frame $B(a,z)$ and then performs a rank-one exactification to restore the telescoping identity.

4.1 Edge slope frame

Let $e^{(j)}$ be the point obtained from a by replacing only the j-th coordinate with z_j. Define

$$B_{ij}(a,z) = \begin{cases} \dfrac{F_i(e^{(j)}) - F_i(a)}{z_j - a_j}, & z_j \neq a_j, \\ 0, & z_j = a_j. \end{cases}$$

This matrix is cheap and locally informative, but it does not generally satisfy (1).

4.2 Rank-one exactification

Let

$$r(a,z) = F(z) - F(a) - B(a,z)(z-a)$$

be the defect of the edge frame. Choose a covector $w(a,z) \in (\mathbb{C}^n)^*$ such that $w(a,z)(z-a) \neq 0$. Define

$$Q^{\mathrm{sys}}(a,z) = B(a,z) + \frac{r(a,z)\, w(a,z)}{w(a,z)(z-a)}. \tag{4}$$

Proposition 4.1 (Exactification)**.** *The matrix Q^{sys} defined by (4) satisfies*

$$F(z) - F(a) = Q^{\mathrm{sys}}(a,z)(z-a).$$

Proof. Multiplying (4) by $z - a$ gives

$$Q^{\mathrm{sys}}(a,z)(z-a) = B(a,z)(z-a) + r(a,z) = F(z) - F(a).$$

□

The adaptive covector. In the implementation, $w(a,z)$ is chosen from the actual system: support activity, coefficient magnitudes, local monomial energy on the box joining a and z, midpoint Jacobian column energy, and residual alignment. This does not change the proof of Proposition 4.1; it changes only the conditioning of the resulting matrix.

4.3 Blended geometries

If Q_1 and Q_2 both satisfy (1), then so does

$$Q_\theta = \theta Q_1 + (1-\theta) Q_2 \qquad (0 \leq \theta \leq 1).$$

Flow 092 inherits the mode portfolio

system, integral_gl, blend.

The blend mode uses the exactified system geometry and the integral geometry. It is still an exact Pandrosion geometry because exactness is affine.

5 Local acceleration and damping

For a fixed anchor a, define

$$H(z) = \mathcal{P}_Q(a,z).$$

The implemented corrector first tries the plain Pandrosion step $H(z)$. If the residual decreases sufficiently, it attempts a coordinatewise Aitken acceleration.

Definition 5.1 (Pandrosion T_2 step)**.** Let

$$s_1 = H(z), \qquad s_2 = H(s_1).$$

The T_2 candidate is the vector whose k-th component is

$$(T_2(z))_k = z_k - \frac{(s_{1,k} - z_k)^2}{s_{2,k} - 2s_{1,k} + z_k}, \tag{5}$$

with the convention that $s_{2,k}$ is used when the denominator is too small.

The step is guarded by two rules:

- if the accelerated point is non-finite or too far from the current scale, fall back to s_2;
- accept only a damped point

  $$z + \tau(c - z), \qquad \tau \in \{1, 0.72, 0.5, 0.35, 0.25, 0.125, 0.0625\},$$

  that reduces the residual.

Remark 5.2 (Why T_2, not T_3 or T_4)*.* The repository contains an experimental wrapper comparing T_2, T_3, T_4. In the tested regimes, T_3 and T_4 sometimes reduce the number of candidate paths but generally increase wall-clock time, and T_4 can destabilize dense bivariate cases. Thus Flow 092 uses T_2 as the default local acceleration.

6 Homothetic transport

The scalar paper uses scaling to replace a large ratio x by a reduced ratio near 1. In several variables, Flow 076 implements the analogue as a diagonal homothety.

Definition 6.1 (Diagonal homothety)**.** Let $S = \mathrm{diag}(s_1, \dots, s_n)$ be an invertible diagonal matrix and let $D = \mathrm{diag}(d_1, \dots, d_n)$ be an optional equation normalization. Define

$$G(y) = D\,F(Sy), \qquad z = Sy.$$

Proposition 6.2 (Transport of Pandrosion slopes)**.** *Assume $Q_G(a, y)$ satisfies*

$$G(y) - G(a) = Q_G(a, y)(y - a).$$

Then

$$Q_F(Sa, Sy) = D^{-1} Q_G(a, y) S^{-1}$$

satisfies

$$F(Sy) - F(Sa) = Q_F(Sa, Sy)(Sy - Sa).$$

Proof. Since $G(y) = DF(Sy)$,

$$D(F(Sy) - F(Sa)) = Q_G(a, y)(y - a).$$

Multiplying by D^{-1} and using $y - a = S^{-1}(Sy - Sa)$ gives the claim. □

The scales S are generated from coefficient and support balancing, and in the bivariate case can be blended with a Pandrosion polygon/Jensen estimate. Equation normalization D equalizes component magnitudes. The key point is that scaling changes the geometry of the path without breaking (1).

7 Total-degree homotopy in $\mathbb{C}^n$

Flow 092 is still a homotopy method. It uses total-degree starts

$$G_i(z) = z_i^{d_i} - \phi_i,$$

where the phases $\phi_i \in \mathbb{C}^\times$ are deterministic complex phases. The start roots are the Cartesian product of the roots of the G_i. Their number is exactly $D = \prod_i d_i$.

For a vector of nonzero complex phases $\gamma = (\gamma_1, \dots, \gamma_n)$, write

$$(\Gamma F)_i = \gamma_i F_i.$$

The tracked homotopy is

$$H_t(z) = (1-t)G(z) + t\,\Gamma F(z), \qquad 0 \le t \le 1. \tag{6}$$

Predictor. The tracker uses a tangent predictor when stable. Formally, differentiating the path identity $H_t(z(t)) = 0$ gives

$$J_{H_t}(z)\dot{z} + \partial_t H_t(z) = 0.$$

The predictor estimates $z(t + \Delta t)$ from this relation or from a secant history.

Corrector. At the predicted point, the corrector is not a Newton–Lairez corrector. It is the Pandrosion corrector of Sections 2–5 applied to the system $H_{t+\Delta t}$.

Branch safety. A corrected point is accepted only if:

$$\|H_{t+\Delta t}(z_{\mathrm{corr}})\| \quad \text{is small}$$

and the correction is compatible with the predictor move. If the correction is too large relative to the local scale or predictor displacement, the step is declared ambiguous and the time step is reduced. This is the purpose of the branch-safe tracker in Flow 087.

8 Bivariate coordinate resultants

For $n = 2$, Flow 089 and Flow 092 use coordinate resultants as recovery devices. Let

$$F = (F_1, F_2) \in \mathbb{C}[x, y]^2.$$

Define

$$R_x(x) = \mathrm{Res}_y(F_1(x,y), F_2(x,y)), \qquad R_y(y) = \mathrm{Res}_x(F_1(x,y), F_2(x,y)). \tag{7}$$

If (ξ, η) is a common zero of F_1, F_2, then

$$R_x(\xi) = 0, \qquad R_y(\eta) = 0.$$

Reconstruction. The implementation samples R_x and R_y on circles and reconstructs their coefficients by a discrete Fourier formula. It then computes their roots with a Durand–Kerner iteration. These roots are not accepted as solutions of the original system. They are used only to generate candidate starts:

$$x = \xi \quad \Rightarrow \quad F_i(\xi, y) = 0$$

for one or both equations, followed by residual ranking and Pandrosion polish on the full system.

Pairing layer. Flow 092 also contains an opt-in layer that cross-pairs roots of R_x and R_y. In the current best default this pair recovery is disabled, because it is often more expensive than the recovery it provides. The more important 092 default is bivariate tight clustering: if the user does not specify a cluster radius and $n = 2$, valid roots are clustered at a much smaller radius than the generic multivariate default.

9 Multivariate and projective completion

When $n \geq 3$, coordinate resultants in the literal multivariate sense are large elimination objects. Flow 092 therefore uses two different completion mechanisms.

9.1 Gamma completion sweeps

Flow 090 reruns deterministic gamma sweeps over the same total-degree start set, but only after the base pass has failed to reach the Bézout count. Each candidate root extracted from these sweeps is accepted only if a Pandrosion polish on the original target system brings the residual below the acceptance threshold.

9.2 Projective/Riemann-sphere charts

Flow 091 treats each coordinate as a Riemann sphere with two affine charts:

$$\text{north: } z_j = y_j, \qquad \text{south: } z_j = \frac{s_j}{y_j}.$$

For a chosen mask of south-chart coordinates, denominators are cleared equation-wise so that the transformed system is again polynomial. A root found in chart coordinates is mapped back to the original z-coordinates and then polished on the original system.

Remark 9.1 (Why projective recovery is late). The projective chart can improve conditioning for roots that are large or badly placed in the affine chart. However, choosing a good chart before seeing any partial root cloud is difficult. The current architecture therefore uses projective/Riemann charts as a recovery tool rather than as the default starting geometry.

10 The Flow 092 algorithm

We can now state the algorithm in mathematical form.

Algorithm 10.1 (Pandrosion dD solver, Flow 092)**.** Input: a square polynomial system $F : \mathbb{C}^n \to \mathbb{C}^n$, degrees d_i, and Bézout target $D = \prod_i d_i$.

1. Build a diagonal homothety $z = Sy$, optionally with equation normalization D_F.
2. Build candidate geometry policies: system-unit geometry in all dimensions, and for $n = 2$ additional polygon/Jensen-informed policies.
3. Score policies on a small sentinel subset of total-degree paths.
4. Select one policy and track total-degree homotopy paths in $\mathbb{C}^n$: predictor by tangent/secant extrapolation, corrector by the Pandrosion portfolio

   $$Q^{\text{sys}}, \quad Q^{\text{int}}, \quad \theta Q^{\text{int}} + (1-\theta)Q^{\text{sys}},$$

   with T_2, damping, and branch guard.
5. If the Bézout count is reached, stop.

6. If $n = 2$, run coordinate-resultant recovery (7).
7. If $n \geq 3$, run multivariate gamma completion sweeps.
8. If roots are still missing, run projective/Riemann-sphere completion.
9. Optionally, run bivariate pair recovery.
10. Return the cluster set of candidates whose original residual is below the acceptance threshold.

Remark 10.2 (Default configuration). The current 092 defaults are tuned for the observed best configuration:

$$\texttt{batch-timeout} = 20, \quad \texttt{parallel-batches} = 4, \quad \texttt{batch-size} = 16, \quad \texttt{micro-batch} = 4,$$

with equation normalization and early stopping at Bézout enabled. Pair recovery is opt-in.

11 Correctness statements

The algorithm is heuristic globally, but several of its internal transformations are exact.

Theorem 11.1 (Geometry exactness)**.** *Every slope matrix used by the main Pandrosion corrector in Flow 092 satisfies the telescoping identity* (1)*, up to quadrature and floating roundoff:*

$$Q^{\text{int}}, \qquad Q^{\text{sys}}, \qquad \theta Q^{\text{int}} + (1-\theta)Q^{\text{sys}}.$$

Proof. For Q^{int}, this is Theorem 3.2. For Q^{sys}, this is Proposition 4.1. The blend is a convex combination of two exact slopes, hence

$$Q_\theta(a,z)(z-a) = \theta(F(z)-F(a)) + (1-\theta)(F(z)-F(a)) = F(z)-F(a).$$

□

Theorem 11.2 (Homothety invariance)**.** *Diagonal scaling and equation normalization preserve the class of Pandrosion slope matrices.*

Proof. This is Proposition 6.2. □

Corollary 11.3 (Local fixed-zero property after scaling)**.** *Let ζ be a regular zero of F. In any scaled coordinate chart $z = Sy$, the corresponding scaled zero $S^{-1}\zeta$ is a fixed point of the scaled Pandrosion map whenever the transported slope matrix is invertible.*

Proof. Combine Proposition 2.3 with Proposition 6.2. □

Remark 11.4 (What is not proved). This paper does not prove that Flow 092 finds all roots of every polynomial system, nor does it claim an average polynomial-time bound. The established average-case theory for polynomial homotopy over $\mathbb{C}$ is due to Beltran–Pardo and Lairez. The purpose here is to present a derivative-free Pandrosion geometry and document a strong empirical solver built from it.

12 Empirical behavior

The following table records representative local runs of the 092 architecture on the repository's synthetic families. Timings are machine-dependent and are included only to indicate scale.

Family	case	Bézout	roots	paths	seconds	status
`sparse_ks`	$(2, 12)$	144	144	201	≈ 4	ok
`dense064`	$(2, 10)$	100	100	110	≈ 7	ok
`ks`	$(2, 12)$	144	144	232	≈ 35	ok
`ks`	$(3, 4)$	64	64	192	≈ 8	ok
`ks`	$(3, 5)$	125	125	250	≈ 20	ok

Negative experiments. Two natural alternatives were tested and rejected as defaults.

- Higher T_n acceleration (T_3, T_4) did not improve wall-clock time reliably over T_2.
- Resultant-first scheduling was inferior to tracking-first scheduling except on very small bivariate cases. Resultants are best used as recovery, not as the main engine.

13 Conclusion

The multidimensional Pandrosion geometry is governed by one simple invariant:

$$F(z) - F(a) = Q(a, z)(z - a).$$

The segment-integral slope Q^{int} proves that this invariant is canonical in any dimension; the system-adaptive exactification shows how to construct cheaper geometries without losing exactness; homothetic transport preserves the identity under scaling; and the branch-safe homotopy tracker uses these slopes as practical correctors over $\mathbb{C}^n$.

Flow 092 is therefore not merely a collection of numerical tricks. It is a layered implementation of a coherent geometric principle:

$$\begin{aligned}&\text{exact two-point geometry} + \text{homothetic conditioning}\\&\qquad\qquad + \text{guarded continuation} + \text{Pandrosion-certified recovery.}\end{aligned}$$

The current evidence suggests that this order is essential. The homotopy pass finds the correct basins; resultants, multivariate sweeps, and projective charts repair the missing sheets; and every candidate is ultimately judged by the same Pandrosion polish on the original complex system.

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